GMS GANAPATHI

B.Tech – Computer Science and Engineering
Indian Institute Of Information Technology, D&M, Kancheepuram

+91-9347076225 | ganapathi.gadagamma@gmail.com
GitHub | **LinkedIn** | **Portfolio**

Profile

Computer Science undergraduate with strong foundation in Data Structures and Algorithms. Interested in software development and building practical applications, with exposure to Machine Learning and Agentic AI.

Education

Indian Institute of Information Technology, D&M — Kancheepuram
B.Tech in Computer Science and Engineering; CGPA: 8.05 — *2023 – 2027*

Sri Chaitanya Junior College — Andhra Pradesh
Board of Intermediate Education; Percentage: 97.4% — *2021 – 2023*

Experience

WSAI Summer Intern – CCR5 Inhibitor Design via Mol-LM — On-site
WSAI, under Dr. Nirav Bhatt (Mentor: Mr. Roshan) — *May 2026 – Jul 2026*

- Fine-tuned **NovoMolGen-32M**, a transformer-based molecular language model, using **Augmented Hill-Climb reinforcement learning** to generate novel CCR5 inhibitor candidates for HIV drug discovery.
- Developed a data pipeline to collect 4,000+ known CCR5-related ligands, docked them against the CCR5 receptor, and used the results to seed the RL replay buffer.
- Designed and ran a systematic hyperparameter sweep, identifying a configuration with a competitive docking score (**-13.46 kcal/mol**) while cutting the gradient norm by **9x** and halving the oracle budget versus the baseline.

Projects

DermaGlass – AI Dermatology Platform | *Python, FastAPI, PyTorch, React, LangGraph* | **Live Demo** | **GitHub**

- Built and deployed a full-stack AI healthcare system for automated skin disease diagnosis across 12+ dermatological conditions.
- Trained and fine-tuned **EfficientNet-B0** using a 2-stage transfer learning strategy, achieving **93% validation accuracy**.
- Designed an agentic workflow with **LangGraph**, routing between a diagnosis-report node and a conversational assistant node powered by GPT for personalized, India-centric medical guidance.

Sidekick – Personal AI Assistant | *Python, FastAPI, LangGraph, LangChain, OpenAI API* | **Live Demo** | **GitHub**

- Built and deployed an agentic personal assistant using a **LangGraph worker/evaluator loop** with a bounded retry ceiling for self-evaluating, reliable responses.
- Implemented **Retrieval-Augmented Generation (RAG)** for per-thread file uploads, with chunked document retrieval for context-aware chat.
- Integrated **streaming chat over Server-Sent Events (SSE)**, Tavily web search, and Playwright browser automation with graceful fallback for serverless deployment.

Technical Skills

Languages: Python, C++, C
AI & Machine Learning: Machine Learning, Deep Learning, Agentic AI, Generative AI (RAG, Chatbots)
Frameworks / Libraries: PyTorch, LangChain, LangGraph, Scikit-learn, NumPy, Pandas
Web Development: HTML, CSS, JavaScript, React.js, Flask
Databases: MySQL
Developer Tools: Git, GitHub, VS Code

Achievements

- **Winner** – **Codeathon** (DSA Hackathon), an institute-level competitive programming contest at IIITDM Kancheepuram.
- **Captain** – **IFL** (IIITDM Kancheepuram): Led the team to **2nd place** in the Instititute Football League.

Coursework & Interests

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems
Areas of Interest: Data Science, Agentic AI, Generative AI, Software Development